COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS

# BENCHMARKING MATHEMATICAL REASONING BASED ON VISUAL INPUT IN LARGE LANGUAGE MODELS IN SLOVAK

Master thesis

2026

Lea Gogová

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS

# BENCHMARKING MATHEMATICAL REASONING BASED ON VISUAL INPUT IN LARGE LANGUAGE MODELS IN SLOVAK

Master thesis

| | |
|---|---|
| Study program: | Applied informatics |
| Branch of study: | Applied informatics |
| Department: | Department of Applied Informatics |
| Supervisor: | Mgr. Marek Šuppa |

Bratislava, 2026 Lea Gogová

79997299

Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

## ZADANIE ZÁVEREČNEJ PRÁCE

| | |
|---|---|
| **Meno a priezvisko študenta:** | Bc. Lea Gogová |
| **Študijný program:** | aplikovaná informatika (Jednoodborové štúdium, magisterský II. st., denná forma) |
| **Študijný odbor:** | informatika |
| **Typ záverečnej práce:** | diplomová |
| **Jazyk záverečnej práce:** | anglický |
| **Sekundárny jazyk:** | slovenský |

**Názov:** Benchmarking Mathematical Reasoning Based on Visual Input in Large Language Models in Slovak
*Porovnávacie hodnotenie matematického uvažovania na základe vizuálneho vstupu vo veľkých jazykových modeloch v slovenčine*

**Anotácia:** Nedávne pokroky v oblasti veľkých jazykových modelov (LLM) ukázali ich potenciál pri riešení zložitých úloh zameraných na odôvodňovanie, ktoré sa spoliehajú na vizuálny vstup pre riešenie matematických problémov. Tieto úlohy sa často hodnotia pomocou štandardizovaných testov, ako sú olympiády alebo maturitné skúšky, ktoré vyžadujú nielen počítacie schopnosti, ale aj schopnosť interpretovať diagramy, grafy a vizuálne podnety. Napriek tomu pretrvávajú určité obmedzenia: údaje zo štandardizovaných testov sa často prekrývajú s tréningovými súdbami LLM, čo vedie k skresleným hodnoteniam. Väčšina benchmarkov sa navyše zameriava na angličtinu, pričom v slovenčine, ktorá je jazyk s nízkymi zdrojmi, je dostupných len málo zdrojov. Tento nedostatok venovaných benchmarkov bráni hodnoteniu skutočných schopností LLM v slovenčine a ich používateľnosti v rôznych vzdelávacích kontextoch.

**Cieľ:** Ciele práce zahŕňajú (ale nie sú obmedzené na):
- Preskúmanie, ako LLM pristupujú k matematickému odôvodňovaniu založenému na vizuálnych vstupoch.
- Skúmanie existujúcich benchmarkov na hodnotenie vizuálneho a matematického odôvodňovania.
- Identifikáciu a riešenie medzier v benchmarkoch pre slovenčinu.
- Vypracovanie slovenského datasetu matematických úloh s vizuálnymi komponentmi zo štandardizovaných testov a olympiád.
- Hodnotenie výkonu LLM na datasetoch so zameraním na odôvodňovanie prostredníctvom vizuálnych vstupov.
- Analýzu chýb v odôvodňovaní založenom na vizuálnych úlohách v slovenskom kontexte.

**Literatúra:**
Shi, Freda, et al. "Language models are multilingual chain-of-thought reasoners." arXiv preprint arXiv:2210.03057 (2022). (https://arxiv.org/abs/2210.03057)
Sawada, Tomohiro, et al. "ARB: Advanced reasoning benchmark for large language models." arXiv preprint arXiv:2307.13692 (2023). (https://arxiv.org/pdf/2307.13692.pdf)

79997299

Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

Chen, Nuo et al. "Breaking Language Barriers in Multilingual Mathematical Reasoning: Insights and Observations." ArXiv abs/2310.20246 (2023): n. (https://arxiv.org/abs/2310.20246)
Lu, Pan et al. "MathVista: Evaluating Mathematical Reasoning of Foundation Models in Visual Contexts." International Conference on Learning Representations (2023).

**Vedúci:** Mgr. Marek Šuppa
**Katedra:** FMFI.KAI - Katedra aplikovanej informatiky
**Vedúci katedry:** doc. RNDr. Tatiana Jajcayová, PhD.

**Dátum zadania:** 14.12.2024

**Dátum schválenia:** 15.12.2024 prof. RNDr. Roman Ďurikovič, PhD.
garant študijného programu

študent

vedúci práce

I hereby declare that I have written this thesis by myself, only with help of referenced literature, under the careful supervision of my thesis advisor.

. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .

Bratislava, 2026 Lea Gogová

# Acknowledgement

# Abstract

tba

**Keywords:**

# Abstrakt

tba

**Kľúčové slová:**

# Contents

# List of Figures

# List of Tables

# Motivation

tba

# Chapter 1

# Vision-language models

# Chapter 2

# Reasoning

[Should define what we mean by *reasoning tasks* and when do we say that models *reason*.]

## 2.1 Benchmarks

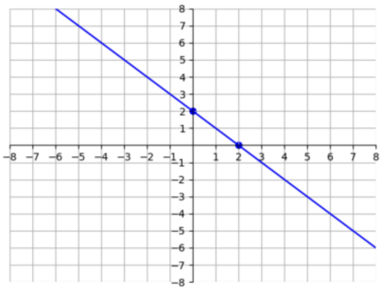
There have been several attempts to create reasonable benchmarks for mathematical reasoning tasks. It all started with evaluation on textual inputs. One of the benchmarks used to this day was **GSM8K**, a collection of elementary school arithmetic word problems. To fill the gap in the realm of more advanced problems, **MATH** dataset comprising Olympiad-level problems was created. OlymMATH is new and should be more challenging. Then some MathEval, which was rejected in openreview. Omni-Math even better. Mention briefly MMLU. But then, sota LLMs have totally failed USAMO high school olympiad questions.

With the rise of multimodal language models, the evaluation has shifted towards assessing mathematical tasks with visual inputs. One of such benchmarks is **Math-Vista** [1], very nice one. Another one is **DynaMath**, but it seems there are only graph-based questions with multiple-seeds, still an interesting one.

## 2.2 Evaluation

[Rule-based evaluation (multi-choice being very popular, but...) vs free-form answers (not very popular...)]

**Q96 from DynaMath**: What is the distance between the x-intercept and y-intercept of blue line?



**Claude-3.5 Answer**:
1. Identify the $x$-intercept and y-intercept: The $x$-intercept is where the line crosses the $x$-axis ($y = 0$). From the graph, this appears to be at $(3, 0)$. The $y$-intercept is where the line crosses the $y$-axis ($x = 0$). From the graph, this appears to be at $(0, 4)$.

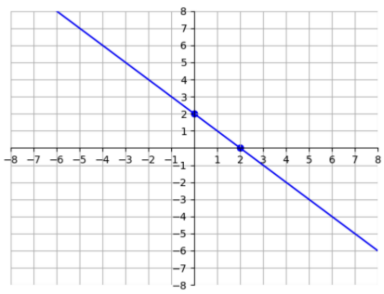
Figure 2.1: Figure reading error.
Btw, ak sme stale len niekde tu, tak to dakujem pekne. Bude to dobra sranda.
Verim, ze vo vlemi blizkej buducnosti taketo somariny niekto vyriesi :D

## 2.3 Prompting techniques

### 2.3.1 Chain-of-thought

### 2.3.2 Cross-lingual

**Q96 from DYNAMATH**: What is the distance between the x-intercept and y-intercept of blue line?



**Claude-3.5 Answer**:
1. Identify the $x$-intercept and y-intercept: The $x$-intercept is where the line crosses the $x$-axis ($y = 0$). From the graph, this appears to be at $(3, 0)$. The $y$-intercept is where the line crosses the $y$-axis ($x = 0$). From the graph, this appears to be at $(0, 4)$.

Figure 2.2: Error classification objection.

Niekedy nie uplne suhlasim s pridelenou error class. A naozaj je niekedy tazke povedat, kde nastava chyba. Napriklad tuto by som ja povedala, ze model len nevie "citat" ten obrazok. Asi si zle pospajal, co je co a na co pozerat (ale to je len moj pocit, ze?). To by ma tiez zaujimalo, ze ako dobre vedia prepajat modely medzi objektmi na obrazku a v texte. Napr. "uvedomuje si", co je vyska, a ako je znazornena jej dlzka na obrazku? Alebo vobec dlzku ktorych stran znazornuju tie cisla na obrazku?

# Chapter 3

# Dataset

The prevailing number of mathematical benchmarks uses only textual descriptions and only a few introduce visual contexts. This disparity makes the reasoning abilities of visual-language models underexplored, especially in low-resource languages like Slovak. To bridge the gap, we introduce a new curated benchmark for assessing visual and mathematical reasoning abilities of multimodal models.

The dataset comprises [tba] textual questions accompanied by images from standardized mathematical exams, namely [tba].

## 3.1 Data collection

This section describes the data collection process. We define the criteria of high-quality visual math reasoning tasks, describe the sources from which they were obtained, and outline the taxonomy with the annotation of our dataset. The final dataset examples can be seen in Figure [tba].

### 3.1.1 Criteria

[What makes for us a good reasoning math task.]

### 3.1.2 Sources

We collect questions from standardized math exams that are held every year in Slovakia. These sources represent a wide range of students, grade levels, difficulty, and thus ensure a diverse coverage of mathematical reasoning tasks.

#### Matura (2000–2025)

The Matura exam is the Slovak high school graduation exam. It is mandatory for students completing secondary education and is used as an entrance criterion for university admission. The math section targets students typically aged 18–19 and covers a wide spectrum of topics including algebra, functions, geometry, statistics, and word problems. The problems require conceptual understanding, procedural knowledge, and occasionally also interpretation of charts and diagrams. We collected a total of **480** word problems from this source.

#### Monitor (2005–2025)

The Monitor exam is a national standardized test taken by ninth-grade students (around age 15) at the end of lower secondary school. It evaluates students' readiness for upper secondary education. The math section includes word problems with arithmetic, basic algebra, geometry, and real-world applications, generally framed in a non-technical language to assess practical mathematical reasoning. We collected **650** problems from this source.

#### Testovanie 5 (2013–2022)

Testovanie 5 is a national standardized test administered to fifth-grade students (around age 11) in elementary school. The exam evaluates basic numeracy, arithmetic operations, geometry, and simple word problems relevant to everyday contexts. The problems are designed to be solvable without advanced symbolic manipulation and emphasize basic reasoning and comprehension. From this source, we gathered **330** problems.

#### Mathematical Olympiad

We also included selected problems from various levels of the Mathematical Olympiad, a nationwide math competition held annually for students across different age groups. The tasks are designed to challenge the best-performing students and are categorized by age (Z5–Z9 for grades 5 to 9, and SŠ for secondary school levels). Olympiad problems are often non-standard and require deeper mathematical insight, multi-step reasoning, and sometimes visual or geometric intuition. Due to the high variability in structure and format, only problems suitable for NLP processing were selected. We extracted **140** carefully curated problems from this source.

### 3.1.3 Taxonomy

Here I want to have some kind of classification of problems based on math topics, math skills needed, nature of image input, and so on...

### 3.1.4 Annotation

[Describe the annotation process if needed]

### 3.1.5 Questions

- What about multi-choice questions, where one choice one image?

## 3.2 Data analysis

# Chapter 4

# Experiments

The chapter describes our evaluation methodology for assessing the visual mathematical reasoning abilities of VLMs on our newly curated benchmark. As the first step, we evaluate [tba] models in total under several prompting techniques. [tba]

## 4.1 Models

We evaluate [tba] models, [tba] of which are open-source, namely...

- Phi4
- GPT-4v

## 4.2 Methods

- baseline
- zero-shot cot
- few-shot cot

## 4.3 Evaluation

# Chapter 5

# Results

In this chapter, we present the results of our experiments.

## 5.1 Limitations

# Bibliography

[1] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts, 2024. `https://arxiv.org/abs/2310.02255`.

[2] Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. Language models are multilingual chain-of-thought reasoners, 2022.